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Chapter 4

THE MEDIUM ACCESS CONTROL SUBLAYER

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2. The Physical Layer
3. The Data Link Layer
4. The Medium Access Control Sub-Layer
5. The Network Layer
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8. Network Security

DLL: Data Link Layer

- LLC: Logical Link Control
- MAC: Medium Access Control

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4.1 The Channel Allocation Problem



➤ Assumptions for Dynamic Channel Allocation -1

✦ Independent Traffic. 独立业务

- The model consists of N independent stations. Once a frame has been generated, the station is blocked and does nothing until the frame has been successfully transmitted.

✦ Single Channel. 单个信道

- A single channel is available for all communication. All stations can transmit on it and all can receive from it.

✦ Observable Collisions. 可检测碰撞

- If two frames are transmitted simultaneously, they overlap in time and the resulting signal is garbled. This event is called a **collision**.
- All stations can detect that a collision has occurred. A collided frame must be transmitted again later.

➤ Assumptions for Dynamic Channel Allocation -2

★ Continuous or Slotted Time. 连续发送 / 时隙发送

- Time may be assumed **continuous**, in which case frame transmission can begin at any instant.
- Alternatively, time may be **slotted** or divided into discrete intervals (called **slots**).
 - Frame transmissions **must** then **begin** at the start of a slot.
- A slot may contain **0**, **1**, or **more** frames, corresponding to an **idle** slot, a **successful** transmission, or a **collision**, respectively.

★ Carrier Sense or No Carrier Sense. 有载波监听 / 无载波监听

- With the **carrier sense** assumption, stations can tell if the channel is in use before trying to use it.
- If there is **no carrier sense**, stations cannot sense the channel before trying to use it.

4.2 Multiple Access Protocols

- ALOHA
 - ✦ Pure ALOHA
 - ✦ Slotted ALOHA
- Carrier Sense Multiple Access Protocols
 - ✦ Persistent and Nonpersistent CSMA
 - ✦ CSMA with Collision Detection
- Collision-Free Protocols
 - ✦ A Bit-Map Protocol
 - ✦ Token Passing
 - ✦ Binary Countdown
- Wireless LAN Protocols

4.2.1 ALOHA



➤ Pure ALOHA

- ★ **frame time**: the amount of **time** needed to transmit the **standard, fixed-length** frame.

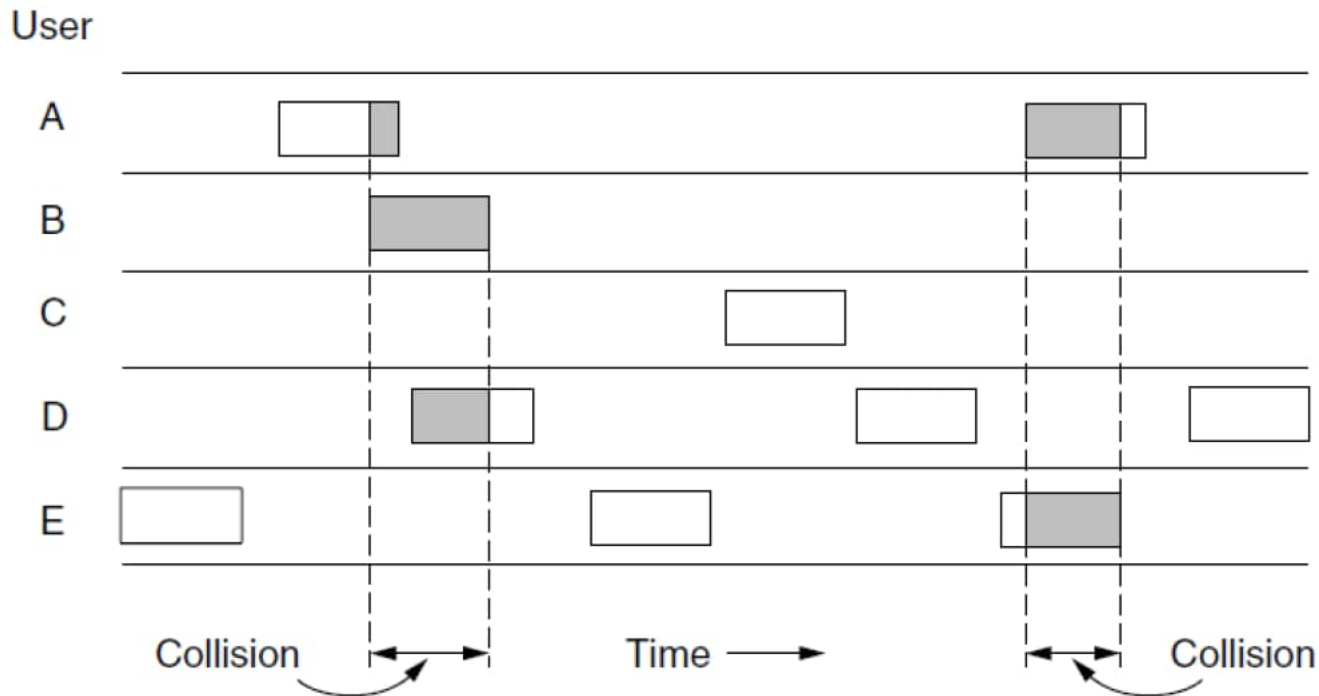
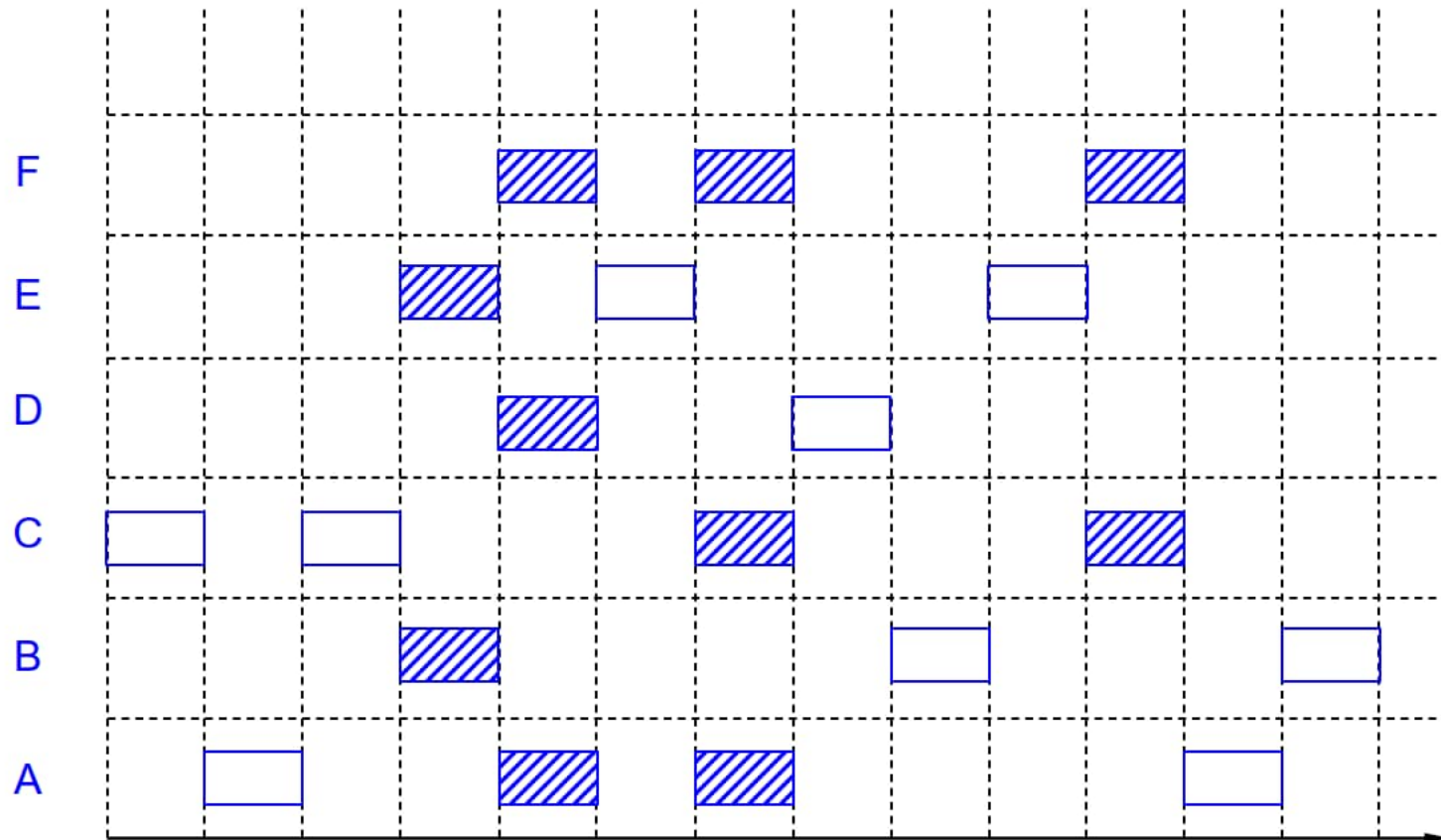


Figure 4-1. In pure ALOHA, frames are transmitted at completely arbitrary times.

Slotted ALOHA



Slotted ALOHA protocol (shaded slots indicate collision)

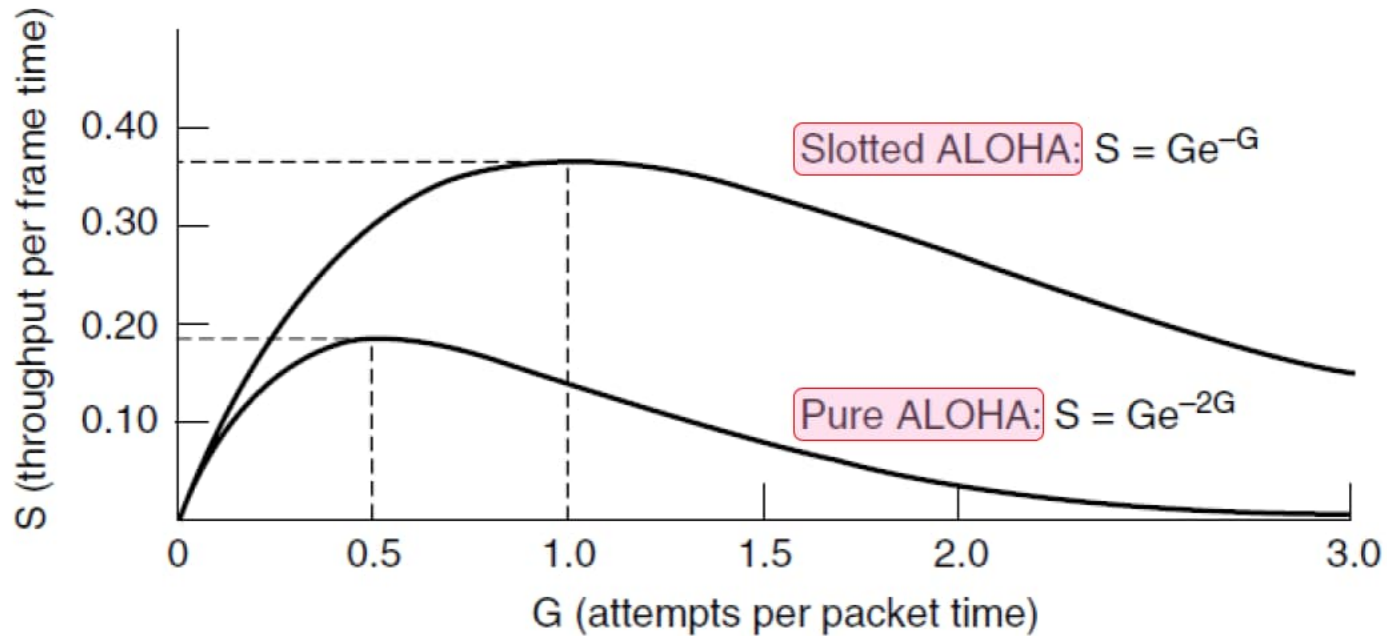


Figure 4-3. Throughput versus offered traffic for ALOHA systems.

4.2.2 Carrier Sense Multiple Access Protocols



- Carrier Sense protocols
 - ✦ Protocols in which stations **listen for a carrier** (i.e., a transmission) and act accordingly.
- CSMA
 - ✦ Carrier Sense Multiple Access 载波侦听多路访问

1-persistent CSMA

- 1-persistent CSMA
 - ✦ first listens to the channel
 - ✦ If the channel is **idle**, the stations **sends** its data.
 - ✦ if the channel is **busy**, the station just **waits** until it becomes idle.
 - Then the station **transmits** a frame.
 - ✦ If a **collision** occurs, the station waits a **random** amount of time and starts all over again.

nonpersistent CSMA



➤ nonpersistent CSMA

- ✦ first listens to the channel
- ✦ If the channel is idle, the stations sends its data.
- ✦ if the channel is already in use,
 - the station does not continually sense it
 - Instead, it waits a random period of time

p -persistent CSMA



➤ p -persistent CSMA.

- ✦ It applies to **slotted** channels
- ✦ When a station becomes ready to send, it **senses** the channel.
- ✦ If it is **idle**, it transmits with a **probability** p .
 - With a probability $q = 1 - p$, it defers until the **next slot**.
- ✦ If that slot is also **idle**, it either transmits or defers again,
 - with probabilities p and q .
- ✦ This process is repeated until either the frame has been transmitted or another station has begun transmitting.
- ✦ If the station initially senses that the channel is **busy**, it waits until the **next slot** and applies the above algorithm.

Comparison of three algorithms



Algorithm	Channel Status		
	Idle (0)	busy (1)	collision (more)
1-persistent	sends data	senses the channel	waits a random amount of time
non-persistent	sends data	waits a random period of time	
p-persistent (for slotted channel)	transmits data with a probability p , defers until the next slot with a probability $(1-p)$		senses the channel

Comparison of the channel utilization

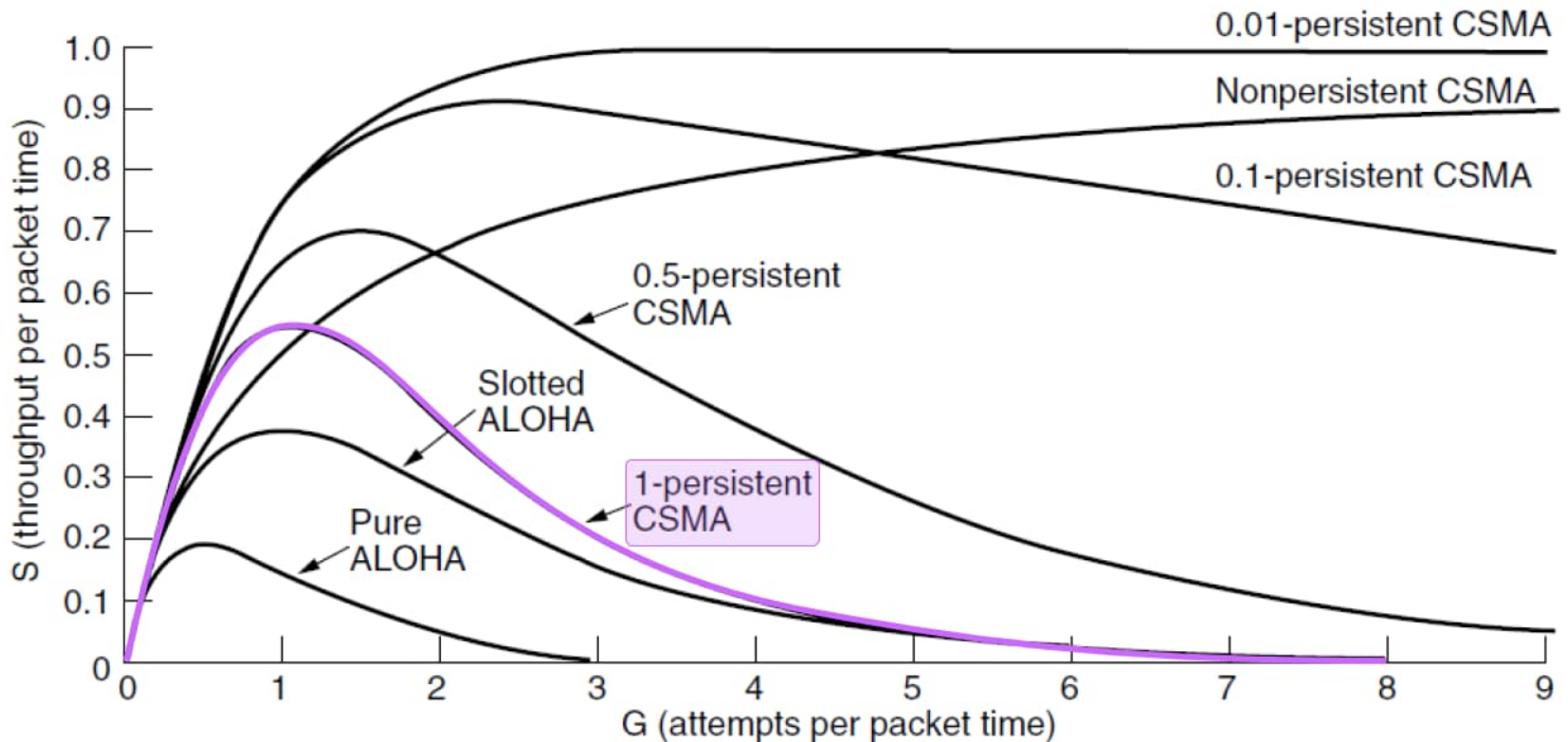


Figure 4-4. Comparison of the channel utilization versus load for various random access protocols.

CSMA with Collision Detection



➤ Collision Detection 碰撞检测

- ★ The station's hardware must listen to the channel while it is transmitting.

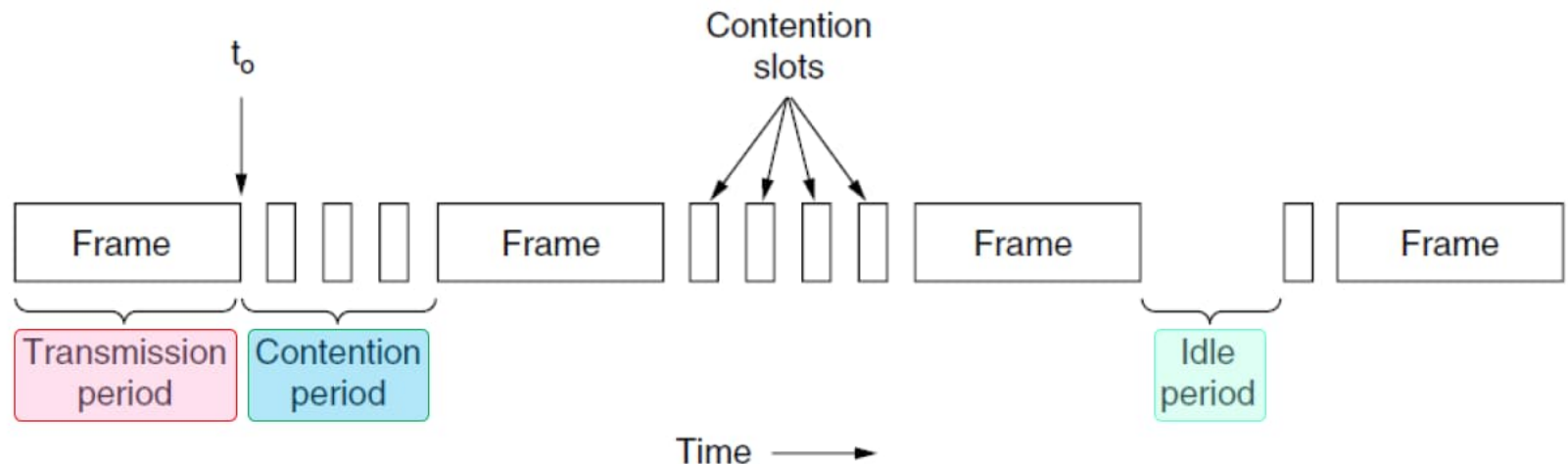


Figure 4-5. CSMA/CD can be in contention, transmission, or idle state.

4.2.3 Collision-Free Protocols

➤ Collision-Free

- ✦ resolve the contention for the channel **without any collisions** at all

➤ we assume that

- ✦ there are exactly N stations, each programmed with a unique address from 0 to $N - 1$.
- ✦ propagation delay is negligible.

basic bit-map method

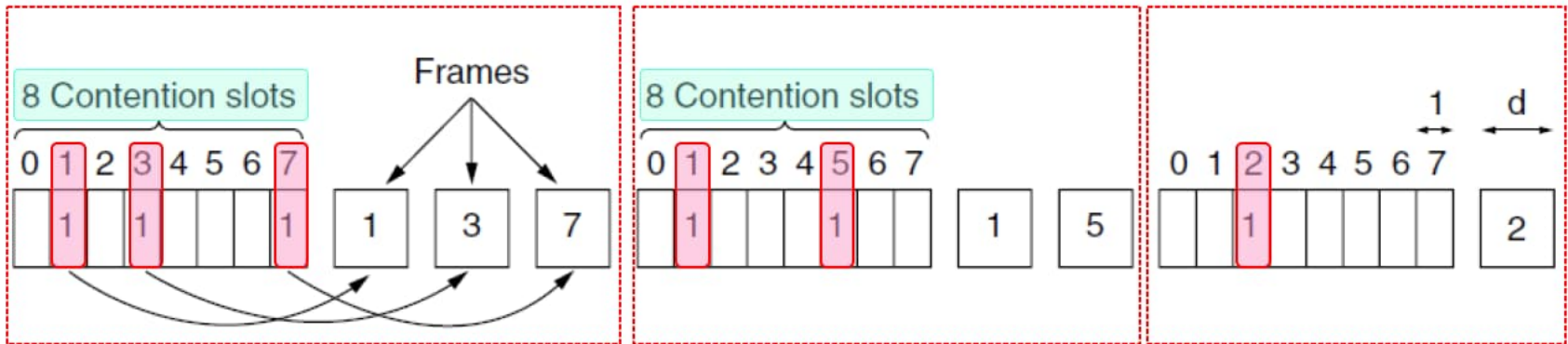


Figure 4-6. The basic bit-map protocol.

Token Passing

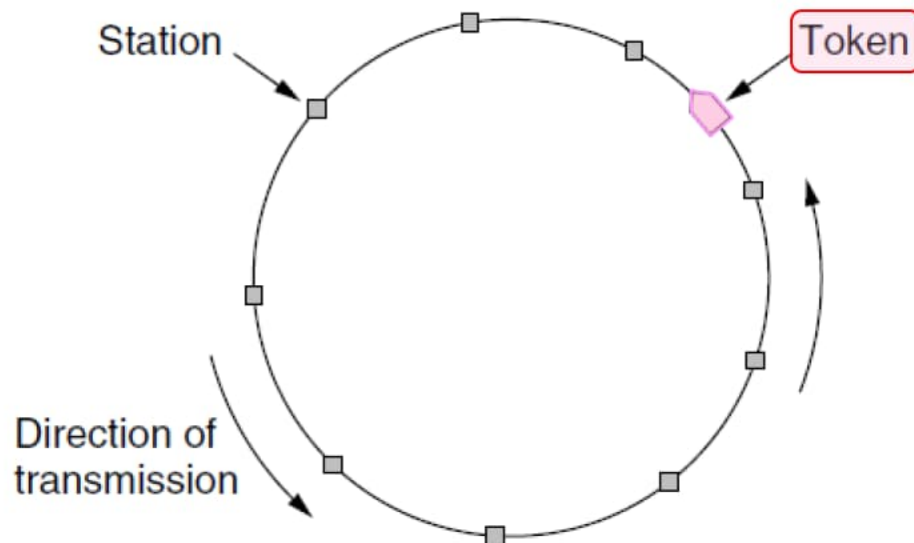


Figure 4-7. Token ring.

Binary Countdown

- Property: higher-numbered stations, have a higher priority.

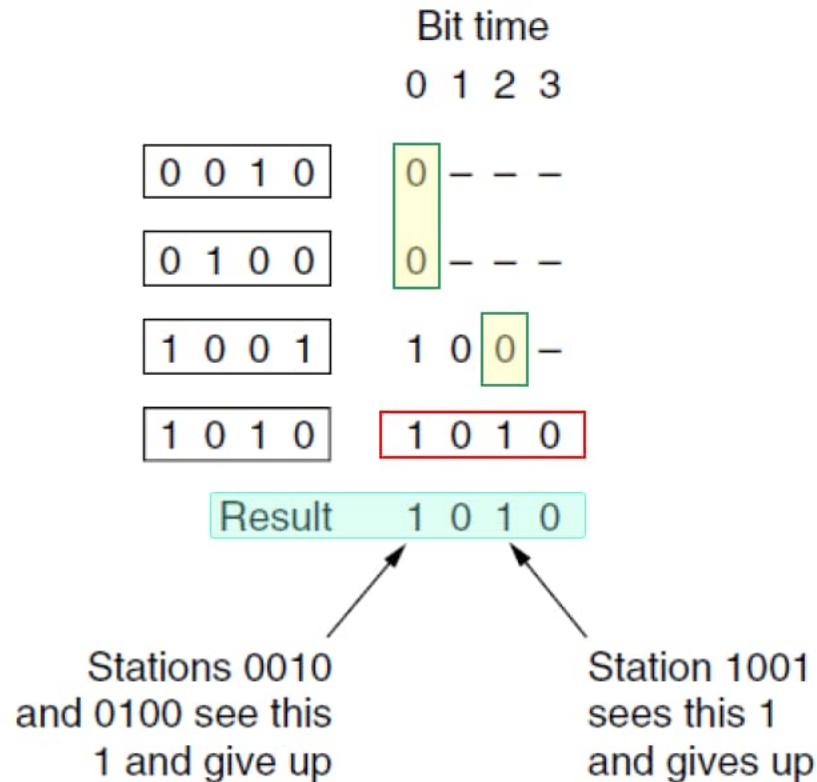
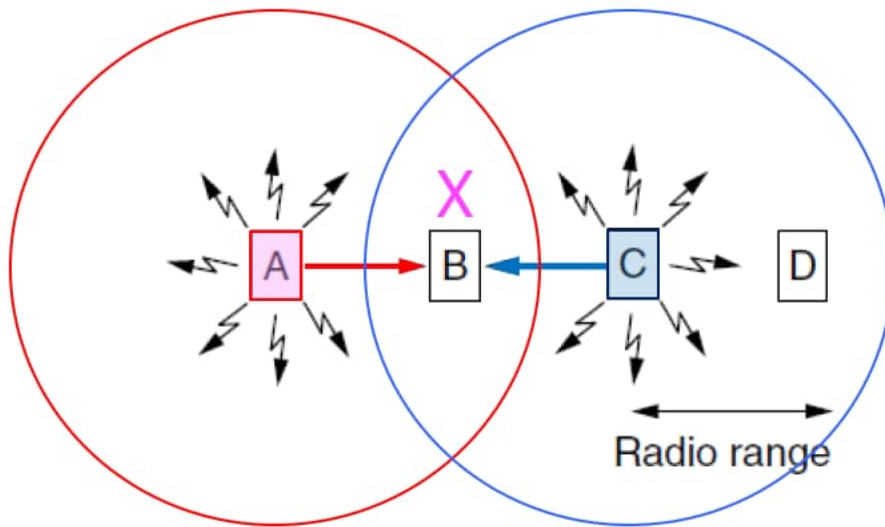
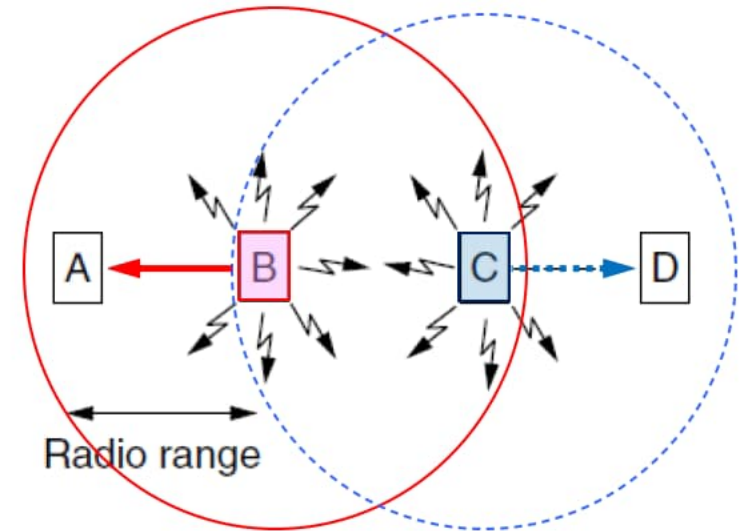


Figure 4-8. The binary countdown protocol.
A dash indicates silence.

4.2.5 Wireless LAN Protocols



(a) A and C are hidden terminals when transmitting to B.



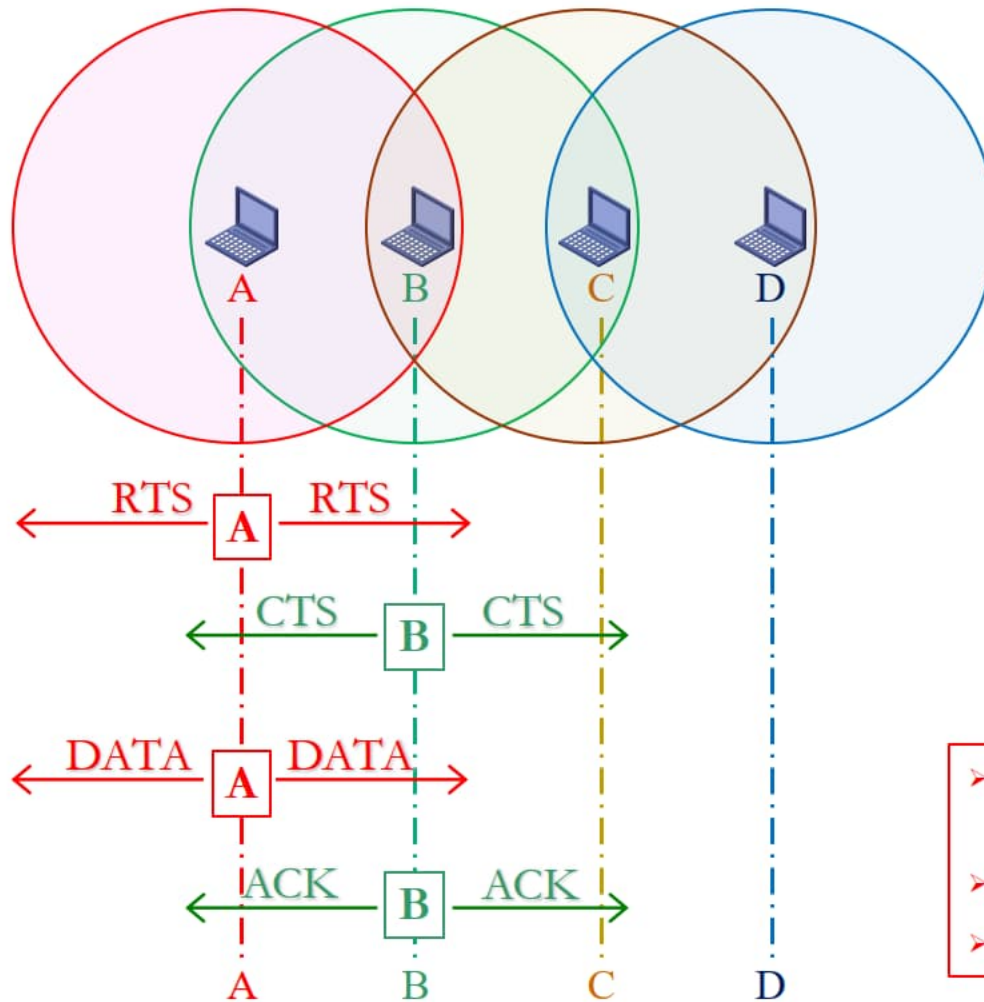
(b) B and C are exposed terminals when transmitting to A and D.

Figure 4-11. A wireless LAN

MACA



多路访问碰撞避免



- MACA (Multiple Access with Collision Avoidance)
- RTS (Request To Send)
- CTS (Clear To Send)

Figure 4-12. The MACA protocol.

4.3 Ethernet



- IEEE 802
 - ✦ 802.3 (Ethernet)
 - ✦ 802.11 (wireless LAN)
 - ✦ 802.15 (Bluetooth, Zigbee)
 - ✦ 802.16 (wireless MAN)
- Two kinds of Ethernet
 - ✦ classic Ethernet
 - ✦ switched Ethernet
 - fast Ethernet
 - gigabit Ethernet
 - 10 gigabit Ethernet

4.3.1 Classic Ethernet Physical Layer



➤ History

- ★ Developed by Bob Metcalfe and others at Xerox PARC in mid-1970s
- ★ Standardized by DIX (DEC, Intel, and Xerox) in 1978
- ★ LAN standards define MAC and physical layer connectivity
 - IEEE 802.3 (CSMA/CD - Ethernet) standard – originally 2Mbps
 - IEEE 802.3u standard for 100Mbps Ethernet
 - IEEE 802.3z standard for 1,000Mbps Ethernet

➤ CSMA/CD: Ethernet's Media Access Control (MAC) policy

- ★ CS = Carrier Sense
 - Send only if medium is idle
- ★ MA = Multiple Access
- ★ CD = Collision Detection
 - Stop sending immediately if collision is detected

Baseband IEEE 802.3



Name	Cable	Max. segment	Nodes/seg.	Advantages
10Base 5	Thick coax	500 m	100	Good for backbones
10Base 2	Thin coax	200 m	30	Cheapest system
10Base-T	Twisted pair	100 m	1024	Easy maintenance
10Base-F	Fiber optics	2000 m	1024	Best between buildings

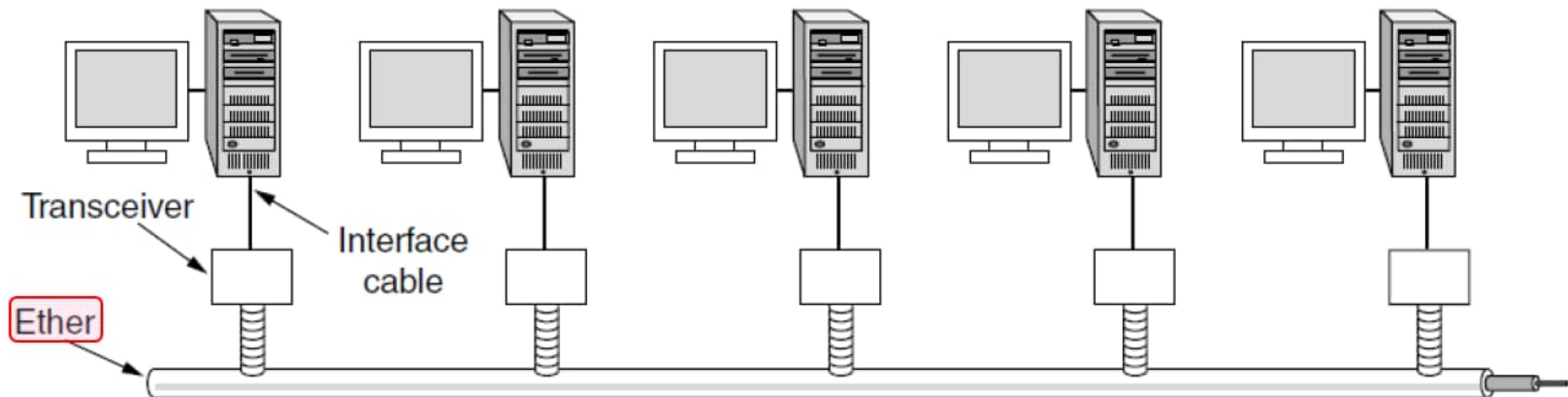
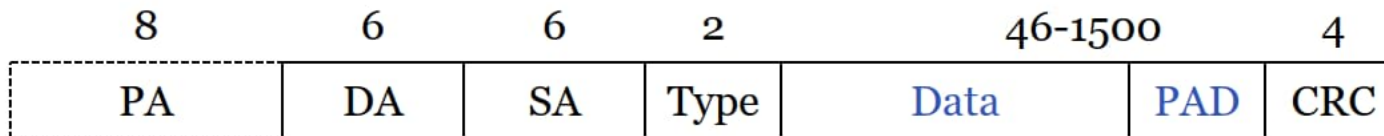
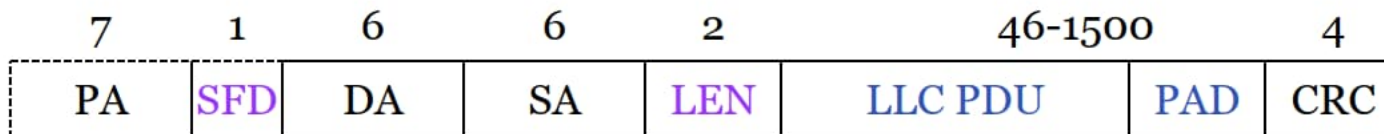


Figure 4-13. Architecture of classic Ethernet.

4.3.2 Classic Ethernet MAC Sublayer Protocol



DIX



IEEE 802.3

PA: Preamble – 10101010s for synchronization

SFD (SoF): Start of Frame Delimiter – 10101011 to start frame

DA: Destination Address – MAC address

SA: Source Address – MAC address

LEN: Length – Number of data bytes < 1500时为长度

Type: identify the higher level protocol > 1500时为类别

LLC PDU + PAD: minimum 46 bytes, maximum 1500

CRC: CRC-32 checksum

Minimal frame length: 64 Bytes = 512 bits
Slot time = 512 bits / 10Mbps = 51.2 μs

Collision detection

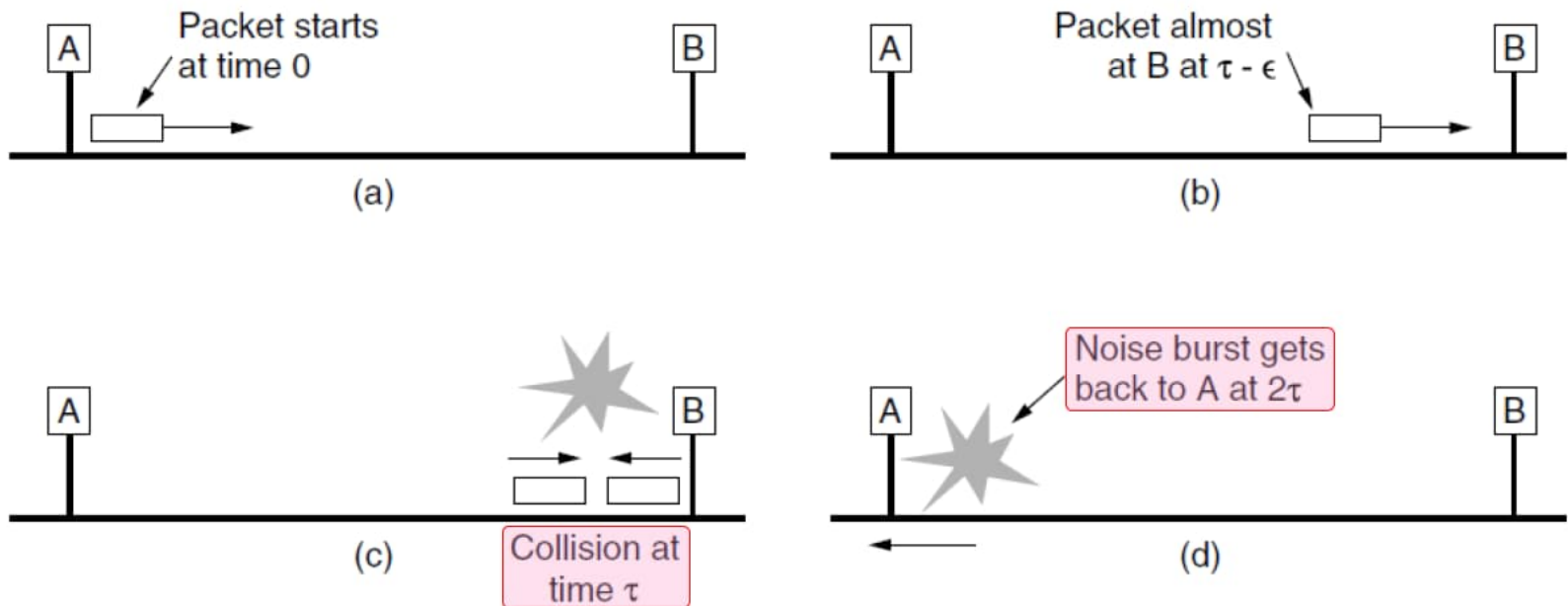
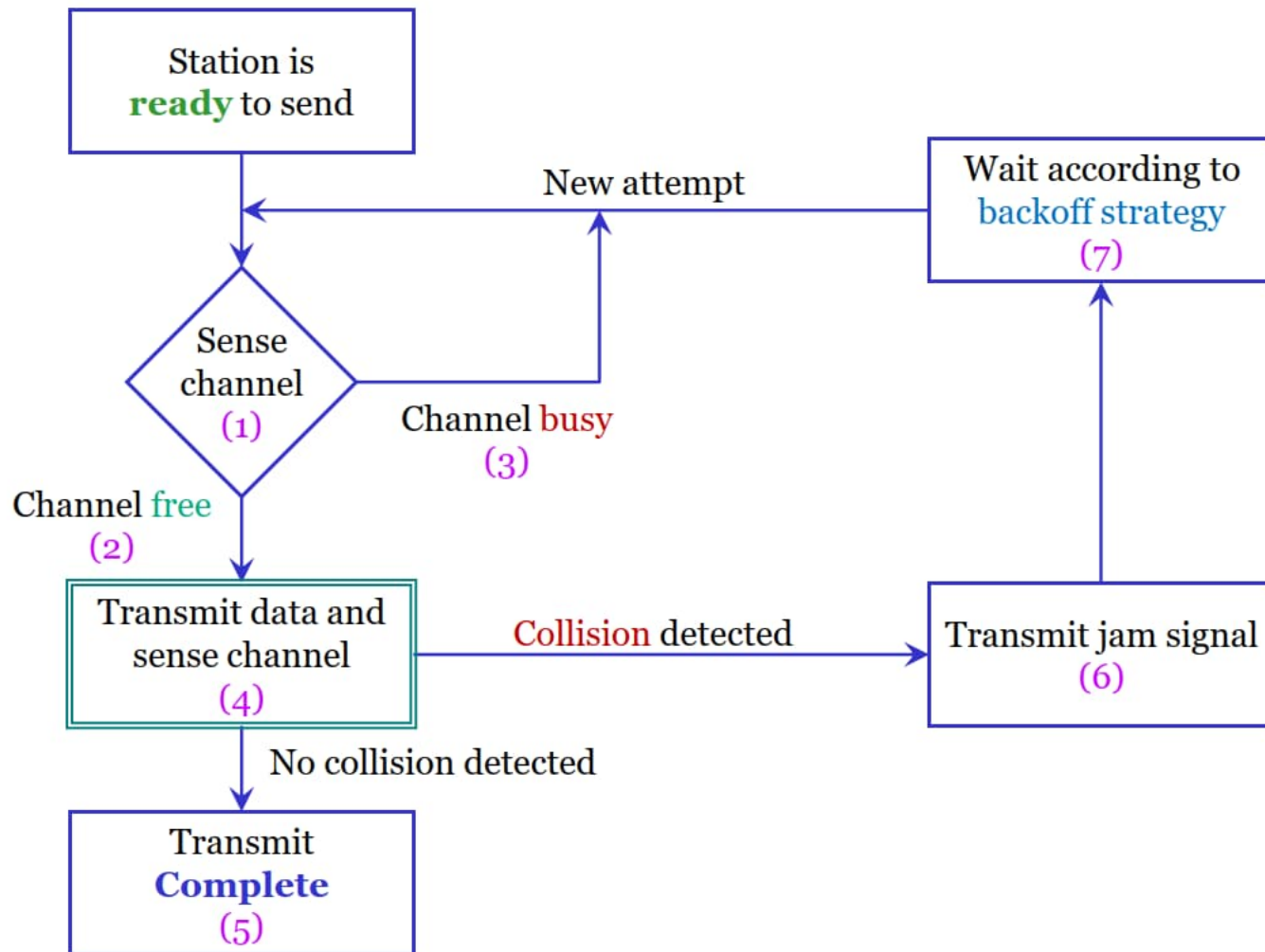
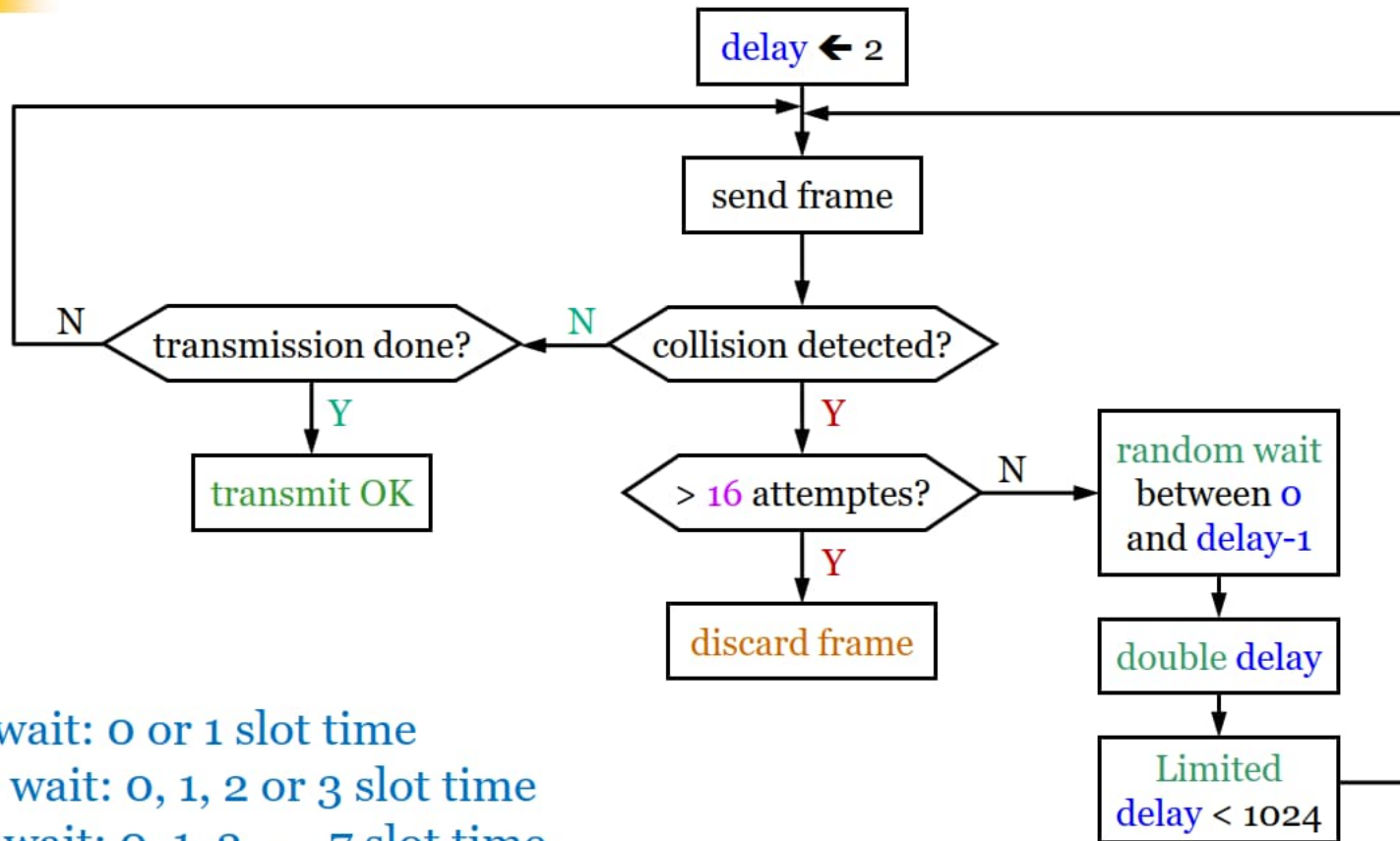


Figure 4-15. Collision detection can take as long as 2τ .

CSMA/CD



Binary Exponential Backoff



1st wait: 0 or 1 slot time
 2nd wait: 0, 1, 2 or 3 slot time
 3rd wait: 0, 1, 2, ..., 7 slot time
 kth wait: 0, ..., (2^k-1) slot time

max delay is limited at
 $1023 * 51.2\mu s = 52.4 \text{ ms}$

- After 10th collisions the interval is frozen at 1023 slots
- After 16th collisions, frame is discarded and report failure back to the upper layer

4.3.3 Ethernet Performance

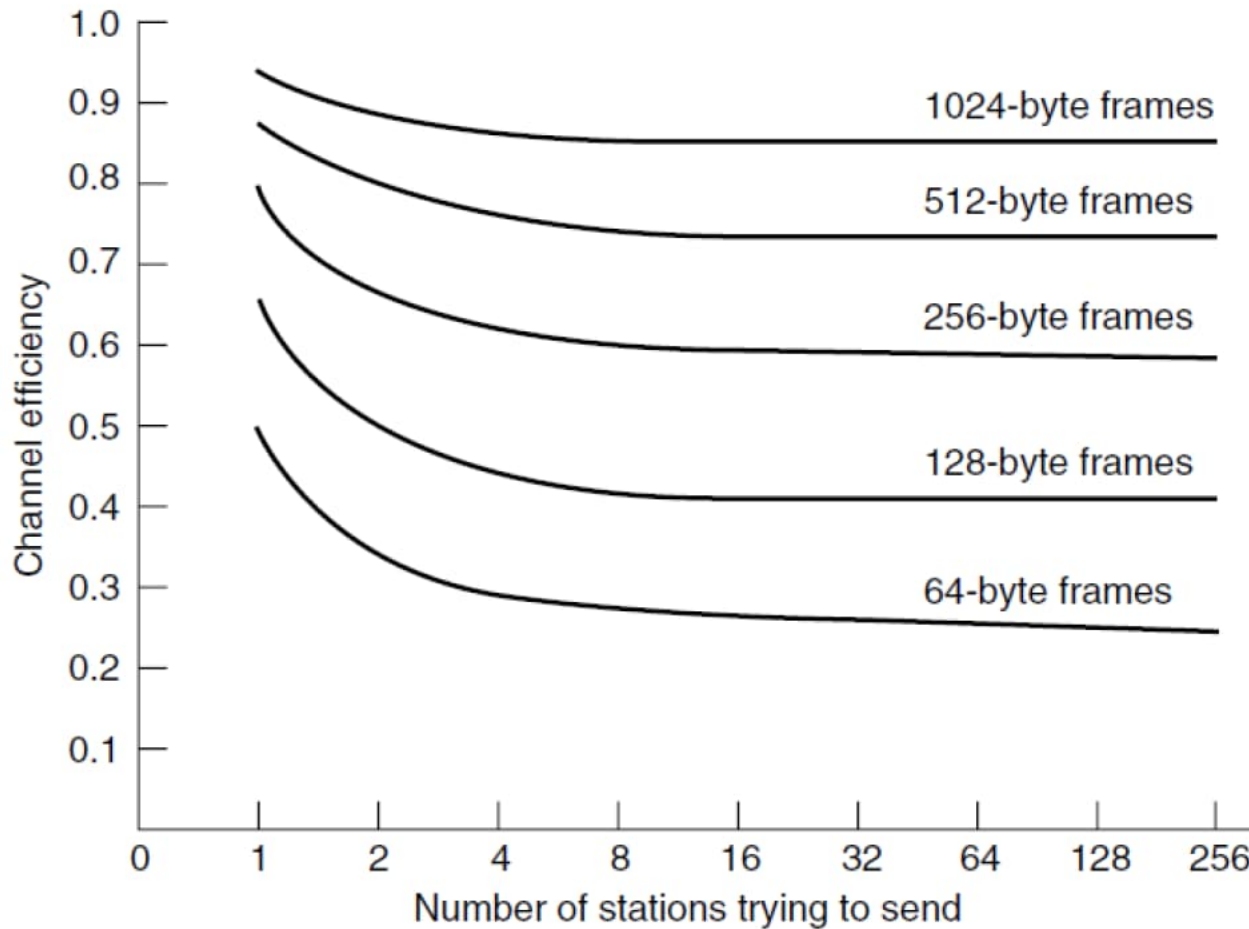
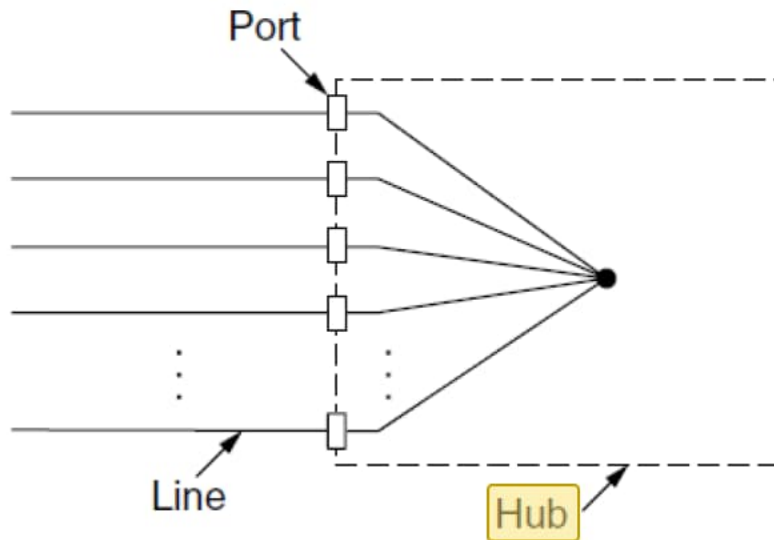


Figure 4-16. Efficiency of Ethernet at 10 Mbps with 512-bit slot times.

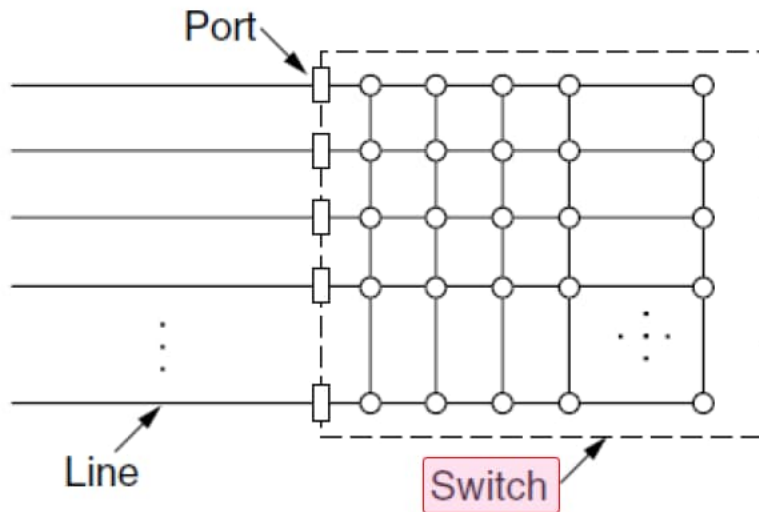
4.3.4 Switched Ethernet



Switched Ethernet: 交换式以太网



(a) Hub.



(b) Switch.

Figure 4-17.

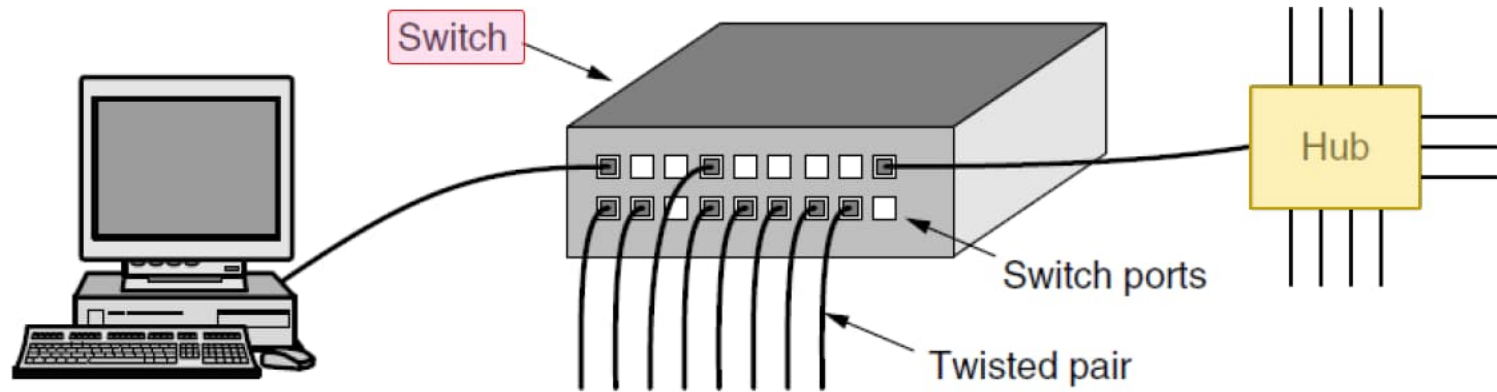


Figure 4-18. An Ethernet switch.

4.3.5 Fast Ethernet



- IEEE 802.3u, in June 1995.

Name	Cable	Max. segment	Advantages
100Base-T4	Twisted pair	100 m	Uses category 3 UTP
100Base-TX	Twisted pair	100 m	Full duplex at 100 Mbps
100Base-FX	Fiber optics	2000 m	Full duplex at 100 Mbps; long runs

Figure 4-19. The original fast Ethernet cabling.

4.3.6 Gigabit Ethernet



- IEEE 802.3^{ab}, in 1999

Name	Cable	Max. segment	Advantages
1000Base-SX	Fiber optics	550 m	Multimode fiber (50, 62.5 microns)
1000Base-LX	Fiber optics	5000 m	Single (10 μ) or multimode (50, 62.5 μ)
1000Base-CX	2 Pairs of STP	25 m	Shielded twisted pair
1000Base-T	4 Pairs of UTP	100 m	Standard category 5 UTP

Figure 4-21. Gigabit Ethernet cabling.

4.3.7 10-Gigabit Ethernet

- for fiber and shielded copper cable appearing
 - ✦ first in 2002 and 2004
- followed by the standard for copper **twisted pair**
 - ✦ in 2006, IEEE 802.3ae

Name	Cable	Max. segment	Advantages
10GBase-SR	Fiber optics	Up to 300 m	Multimode fiber (0.85μ)
10GBase-LR	Fiber optics	10 km	Single-mode fiber (1.3μ)
10GBase-ER	Fiber optics	40 km	Single-mode fiber (1.5μ)
10GBase-CX4	4 Pairs of twinax	15 m	Twinaxial copper
10GBase-T	4 Pairs of UTP	100 m	Category 6a UTP

Figure 4-22. 10-Gigabit Ethernet cabling.

4.4 Wireless LANs

- The main wireless LAN standard is 802.11
 - ✦ IEEE 802.11-2007

4.4.1 The 802.11 Architecture and Protocol Stack

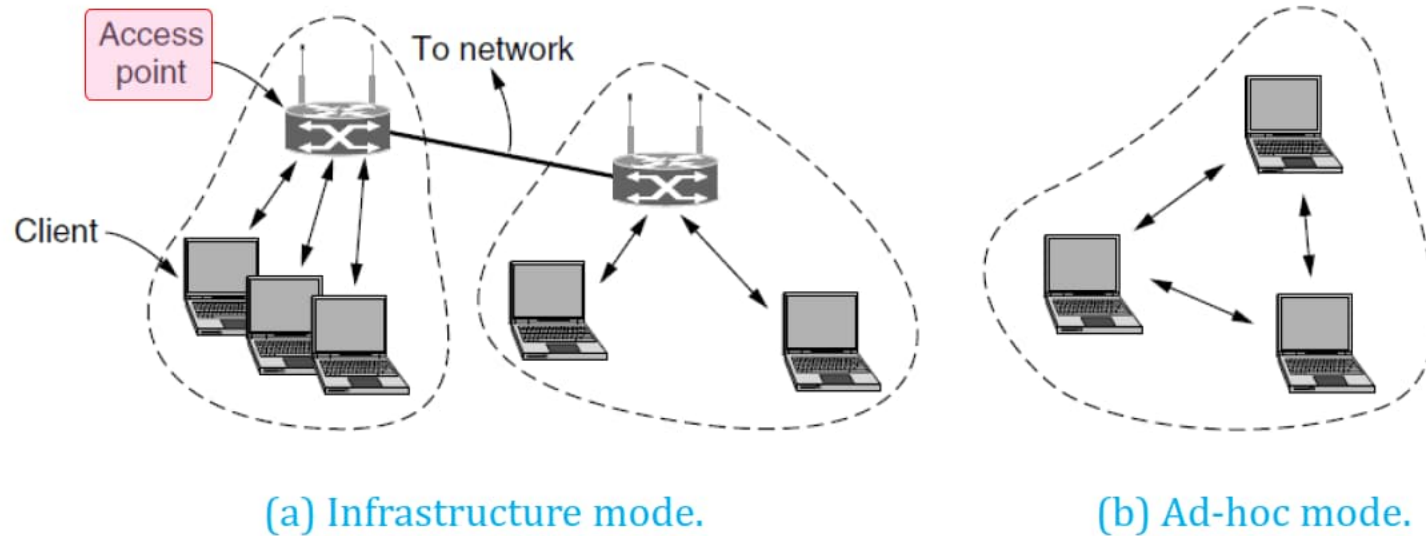


Figure 4-23. 802.11 architecture.

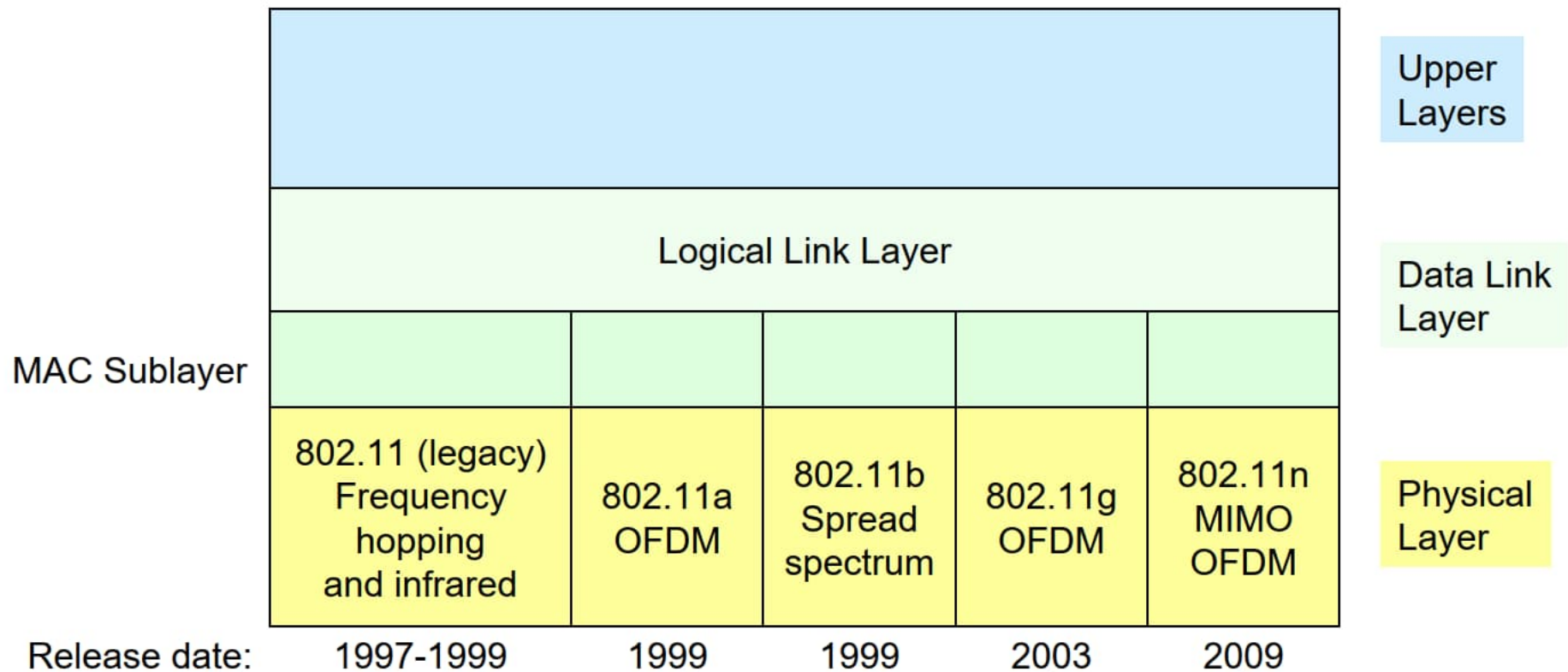
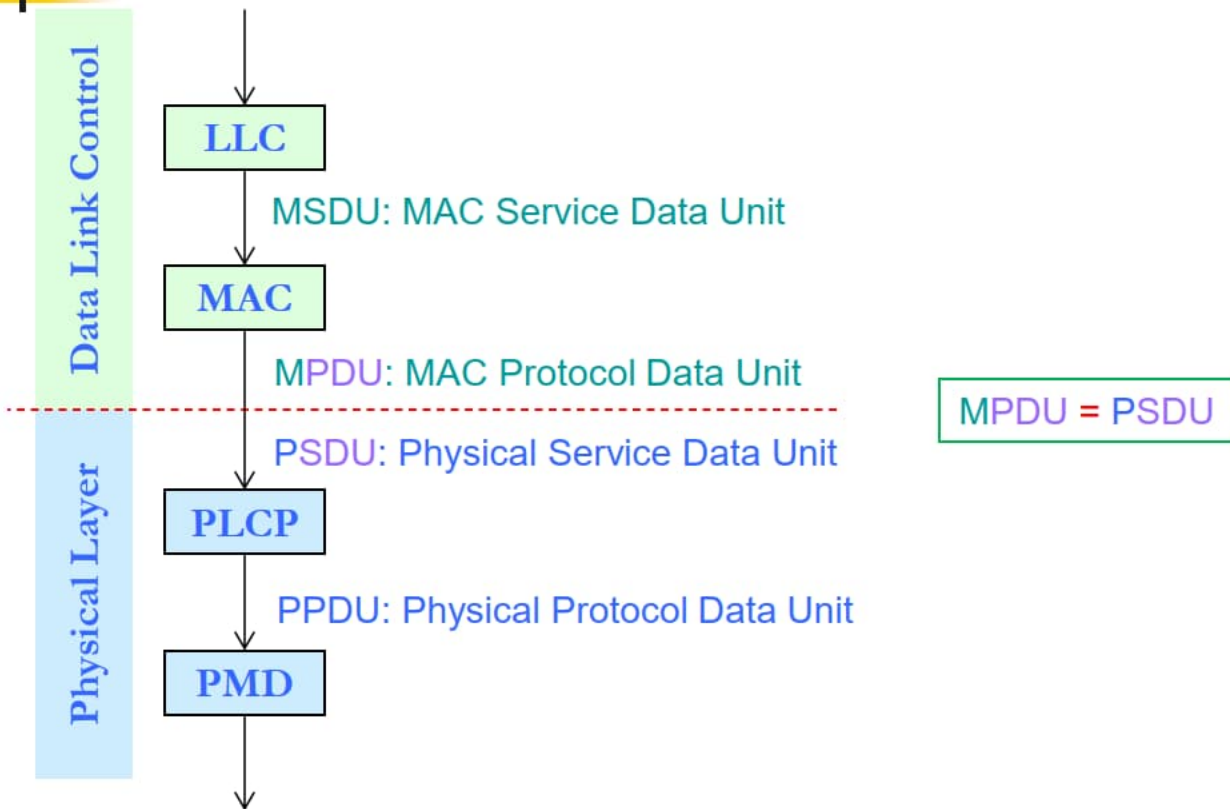


Figure 4-24. Part of the 802.11 protocol stack.



LLC: Logical Link Control 逻辑链路控制 (子层)

MAC: Medium Access Control 介质访问控制 (子层)

PLCP: Physical Layer Convergence Procedure 物理层汇聚处理 (子层)

PMD: Physical Medium Dependent 物理介质相关 (子层)

4.4.2 The 802.11 Physical Layer

- frequency bands
 - ✦ the 2.4-GHz or the 5-GHz ISM
- transmission methods
 - ✦ multiple rates, rate adaptation
 - ✦ 802.11b
 - spread-spectrum method that supports rates of 1, 2, 5.5, and 11 Mbps
 - in the 2.4-GHz ISM band
 - ✦ 802.11a
 - supports rates up to 54 Mbps in the 5-GHz ISM band
 - ✦ 802.11g
 - 6 to 54 Mbps, in the narrow 2.4-GHz ISM band
 - ✦ 802.11n (475Mbps), 802.11ac(1Gbps)

IEEE 802.11 network PHY standards



 Protocol	Release date	Frequency	Bandwidth	Stream data rate	Allowable MIMO streams	Modulation	Approximate range	
		(GHz)	(MHz)	(Mbit/s)			Indoor	Outdoor
802.11-1997	Jun 1997	2.4	22	1, 2	N/A	DSSS, FHSS	20 m	100 m
802.11a	Sep 1999	5	20	6, 9, 12, 18, 24, 36, 48, 54	N/A	OFDM	35 m	120 m
		3.7						5,000 m
802.11b	Sep 1999	2.4	22	1, 2, 5.5, 11	N/A	DSSS	35 m	140 m
802.11g	Jun 2003	2.4	20	6, 9, 12, 18, 24, 36, 48, 54	N/A	OFDM	38 m	140 m
802.11n (Wi-Fi 4)	Oct 2009	2.4/5	20	Up to 288.8	4	MIMO-OFDM	70 m	250 m
			40	Up to 600				
802.11ac (Wi-Fi 5)	Dec 2013	5	20	Up to 346.8	8		35 m	
			40	Up to 800				
			80	Up to 1733.2				
			160	Up to 3466.8				
		0.054–0.79	6–8	Up to 568.9	4			
802.11ad	Dec 2012	60	2,160	Up to 6,757 (6.7 Gbit/s)	N/A	OFDM	3.3 m	
802.11ah	Dec 2016	0.9	1–16	Up to 347	4	MIMO-OFDM		
802.11aj	Est. Jul 2017	45/60						
802.11ax (Wi-Fi 6)	Est. Dec 2018	2.4/5		Up to 10,530 (10.53 Gbit/s)		MIMO-OFDM		
802.11ay	Est. Nov 2019	60	8000	Up to 20,000 (20 Gbit/s)	4	OFDM	10 m	100 m
802.11az	Est. Mar 2021	60						

4.4.3 The 802.11 MAC Sublayer Protocol

➤ CSMA/CA 载波侦听多路访问/碰撞避免

✦ CSMA with Collision Avoidance

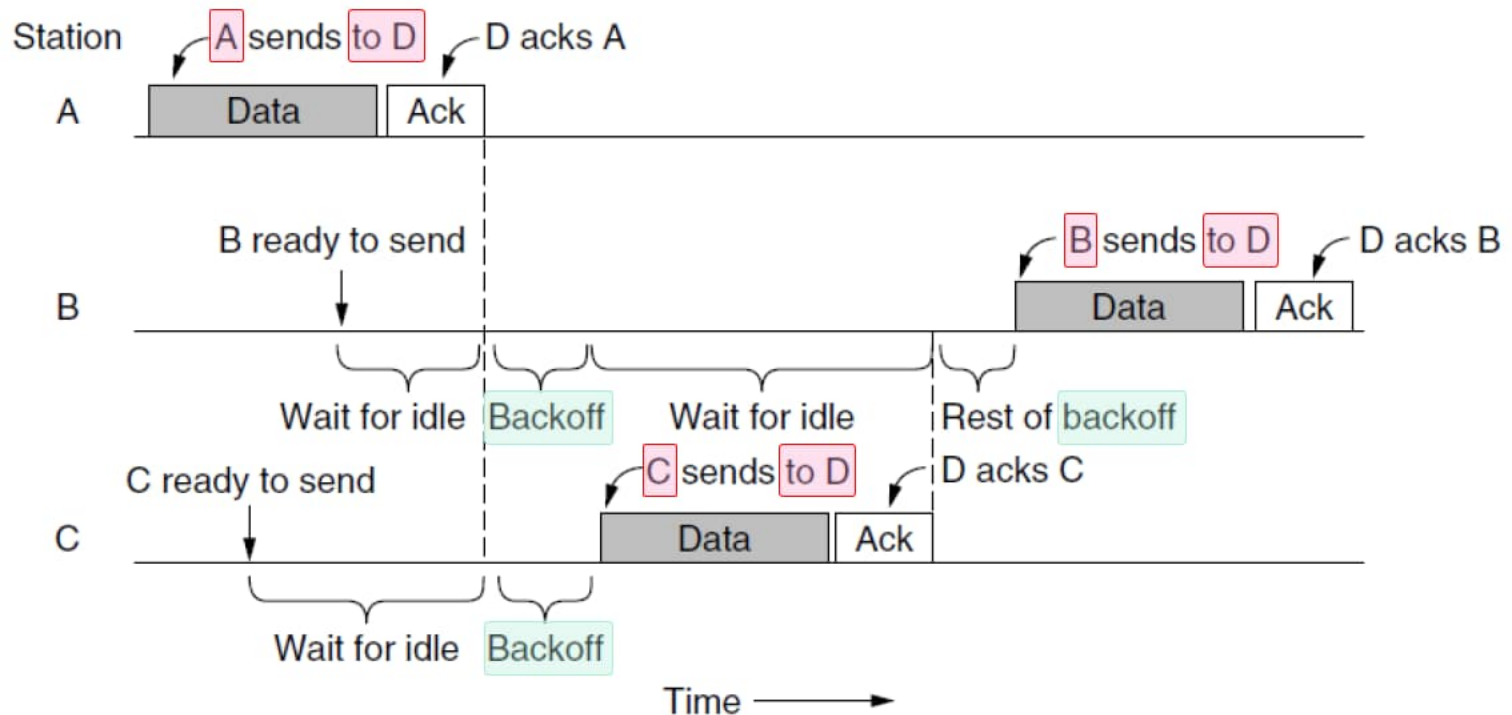
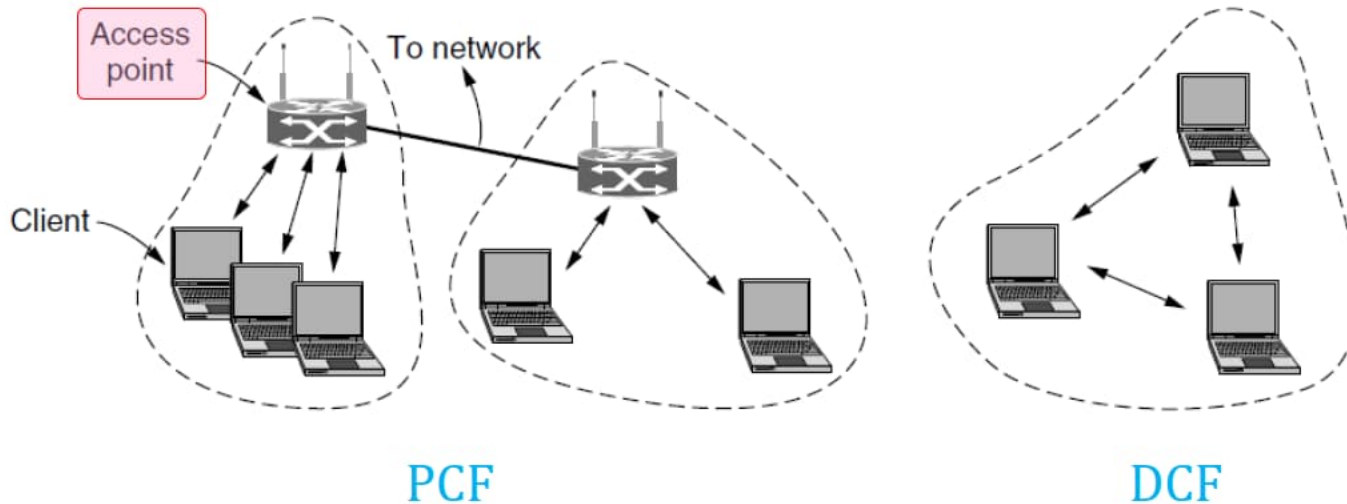


Figure 4-25. Sending a frame with CSMA/CA.

DCF & PCF

- DCF (Distributed Coordination Function) 分布式协调功能
 - ✦ each station acts **independently**, without any kind of central control
- PCF (Point Coordination Function) 点协调功能
 - ✦ the **access point** controls all activity in its cell, just like a cellular base station.



virtual sensing



➤ NAV (Network Allocation Vector) 网络分配矢量

- ✦ Each frame carries a **NAV field** that says **how long** the sequence of which this frame is part will take to complete.
- ✦ Stations that **overhear** this frame know that the channel will be busy for the period indicated by the NAV,
 - regardless of whether they can sense a physical signal.

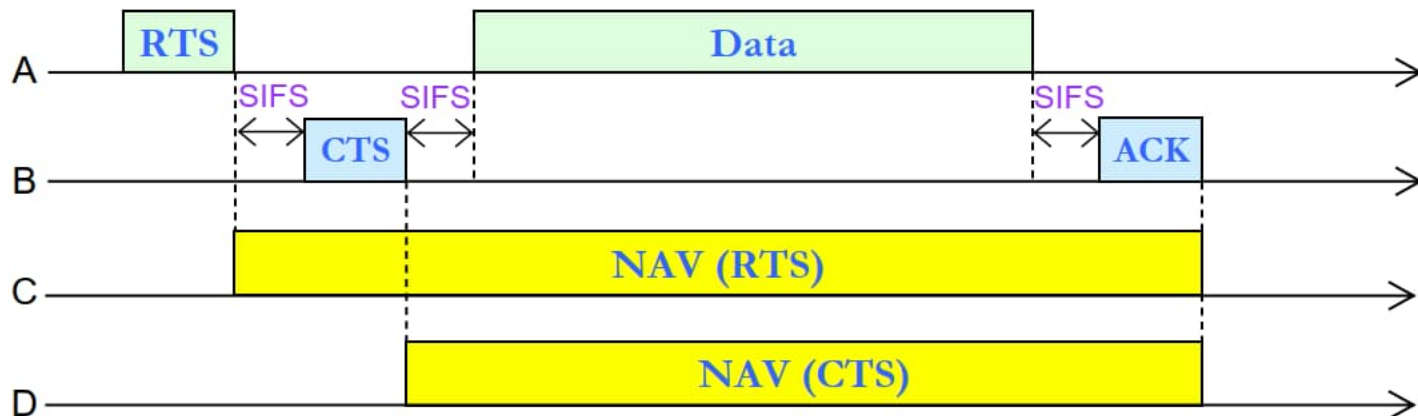


Figure 4-27. Virtual channel sensing using CSMA/CA.

other mechanisms



➤ Reliability 可靠

- ✦ Slower rates use more robust modulations
- ✦ shorter frames
 - fragments: split frames into smaller pieces

➤ Saving Power 节能

- ✦ beacon frames: 信标帧
 - Beacons are periodic broadcasts by the AP (e.g., every 100 msec).
- ✦ power-save mode: 节能模式
 - the client can doze and the AP will buffer traffic intended for it.
- ✦ APSD (Automatic Power Save Delivery): 自动节能递交
 - the AP buffers frames and sends them to a client just after the client sends frames to the AP.
 - The client can then go to sleep until it has more traffic to send (and receive).

InterFrame Spacing

➤ Five intervals

- ✦ SIFS (Short InterFrame Spacing) 短帧间隔
- ✦ AIFS (Arbitration InterFrame Space, for PCF) 仲裁帧间隔
 - AIFS₁(High-priority), AIFS₄(Low-priority)
- ✦ DIFS (DCF InterFrame Spacing) DCF帧间隔
- ✦ EIFS (Extended InterFrame Spacing) 扩展帧间隔

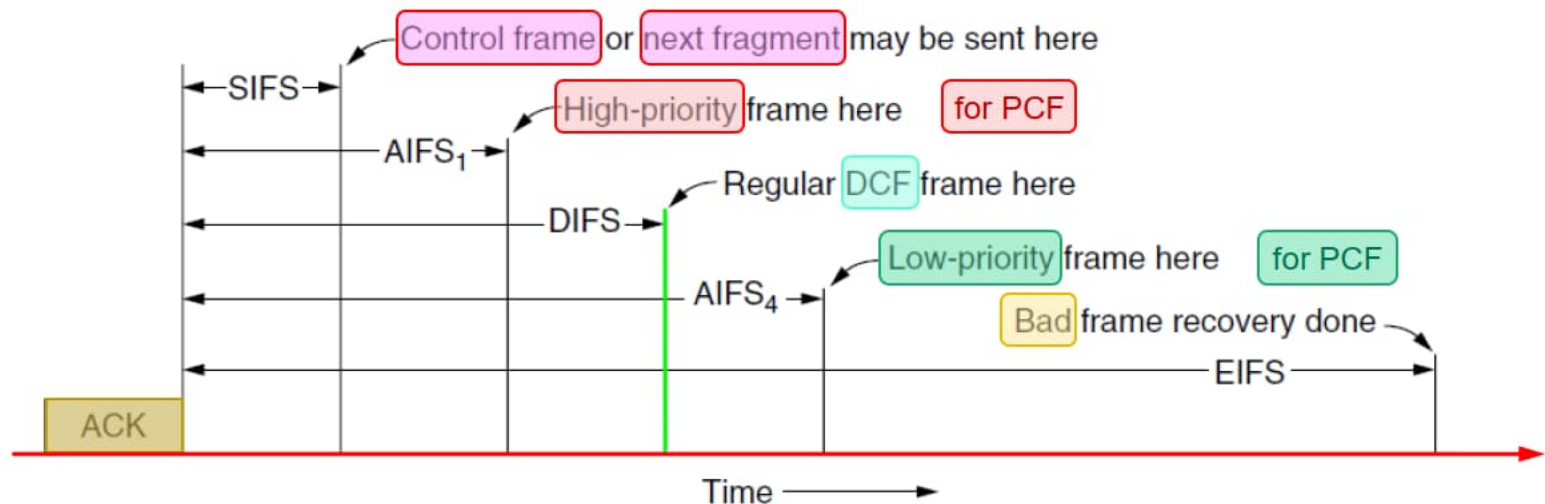
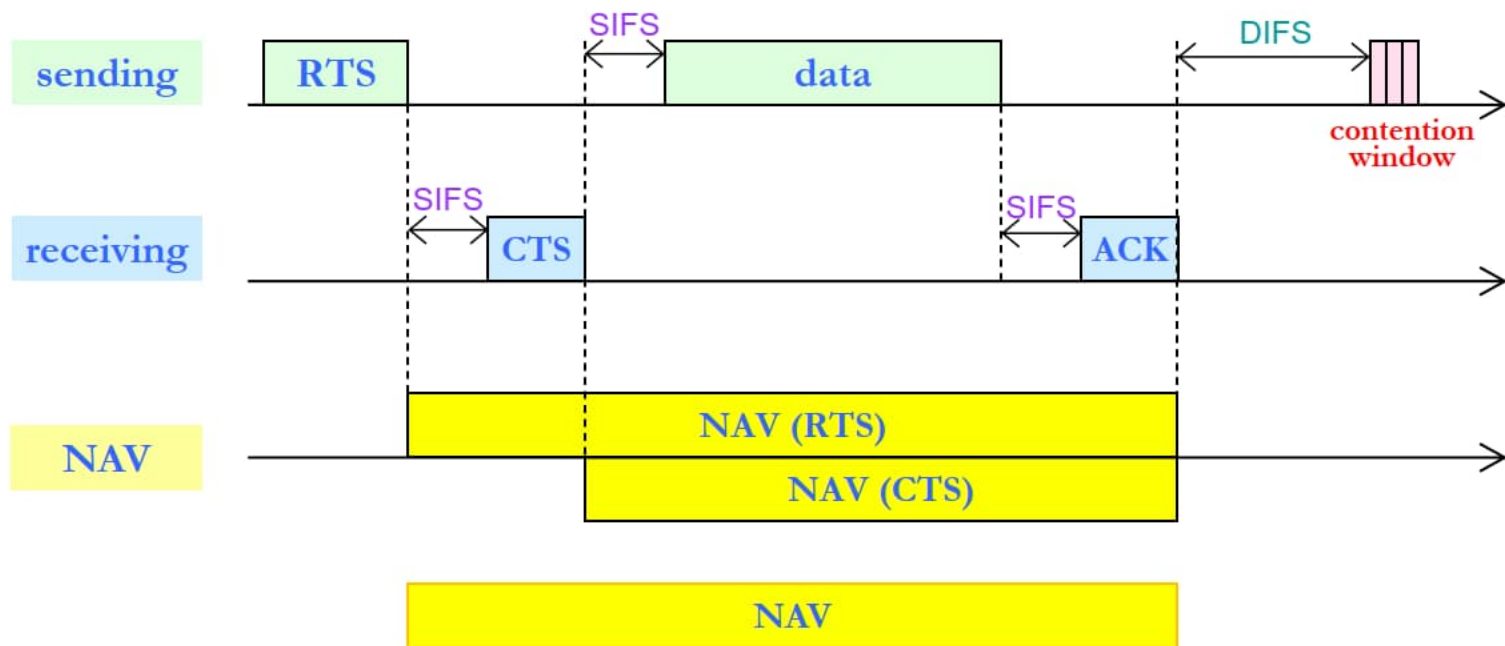


Figure 4-28. Interframe spacing in 802.11.

virtual sensing with NAV



4.4.4 The 802.11 Frame Structure

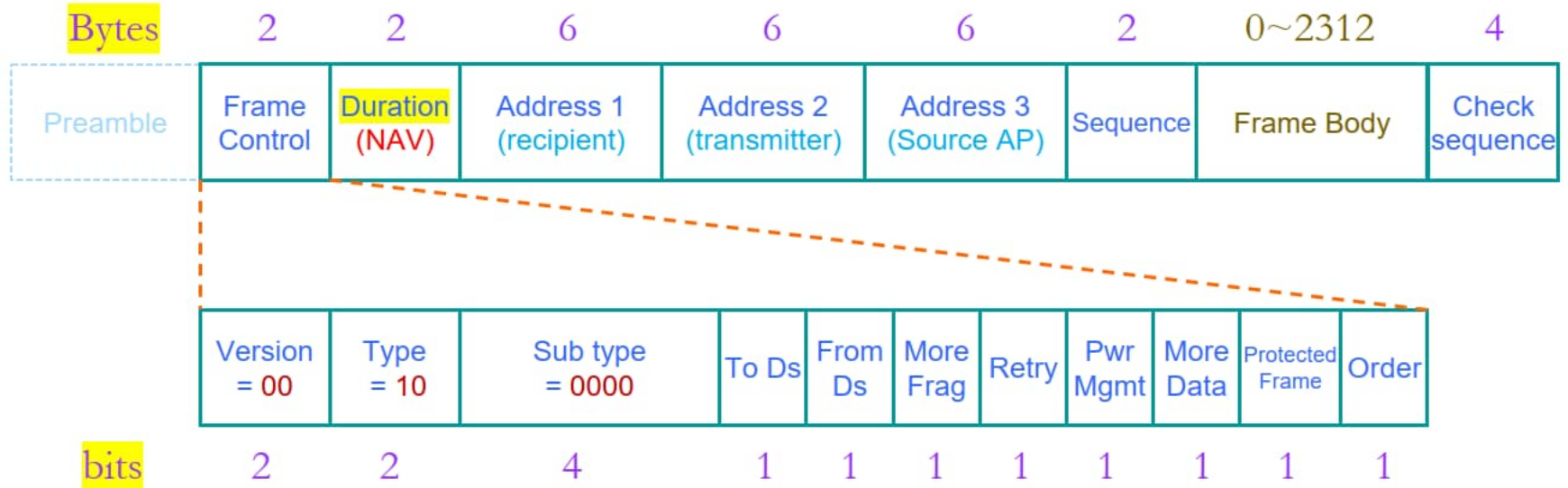


Figure 4-29. Format of the 802.11 data frame.

4.8 Data Link Layer Switching



➤ Bridges 网桥

- ✦ operate in the data link layer
- ✦ examine the data link layer addresses to forward frames
- ✦ provide functionality that goes beyond classic Ethernet and Ethernet hubs to make it easy to join multiple LANs into a larger and faster network.
- ✦ also called “switches” 交换机

4.8.1 Uses of Bridges

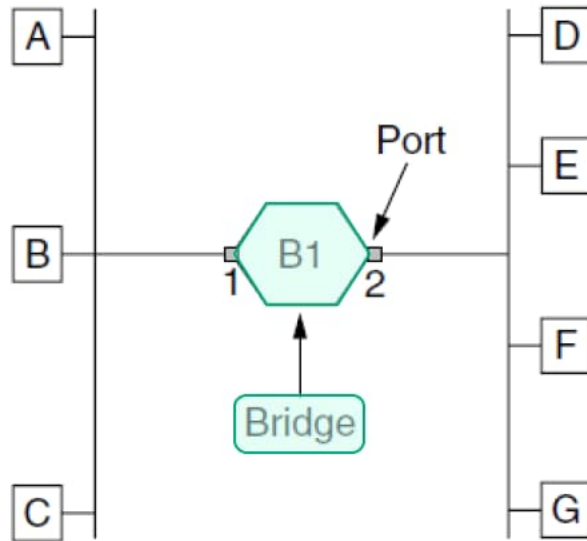


- university and corporate departments
 - ✦ have their own LANs to connect their own personal computers, servers, and devices.
 - ✦ Sooner or later, there is a need for interaction.
- geographically spread over several buildings
- split a single LAN into separate LANs

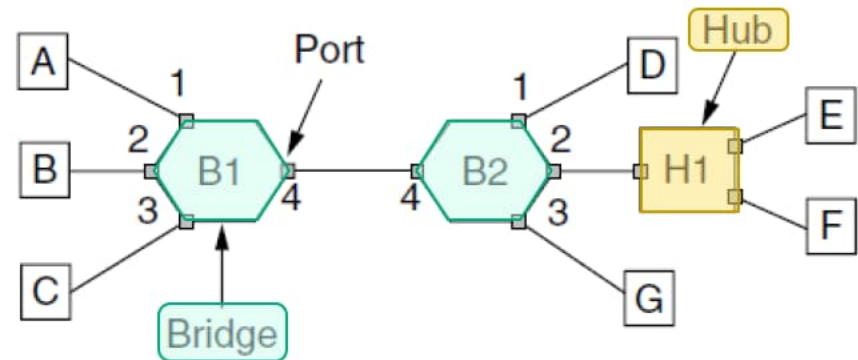
4.8.2 Learning Bridges



Learning Bridges: 学习型网桥



(a) Bridge connecting two multidrop LANs.



(b) Bridges (and a hub) connecting seven point-to-point stations.

Figure 4-41.

multidrop: multi-station

Learning from source & forwarding to destination



基于信源地址的学习，基于信宿地址的转发。

➤ Ethernet switches

- ✦ If the LAN technology is **Ethernet**, the bridges are better known as **Ethernet switches**.

➤ backward learning 反向学习

- ✦ By looking at the **source** addresses, they can tell which machines are accessible on which ports.

➤ routing procedure

1. If the port for the **destination** address is the **same as the source** port, **discard** the frame.
2. If the port for the destination address and the source port are **different**, **forward** the frame on to the destination port.
3. If the destination port is **unknown**, use **flooding** and send the frame on all ports except the source port.

cut-through switching



➤ cut-through switching 直通交换

- ✦ or wormhole routing 虫孔路由
- ✦ Because bridges **only** look at the **MAC addresses** to decide how to forward frames, it is possible to **start forwarding** as soon as the **destination header field** has come in, before the rest of the frame has arrived.
- ✦ usually handled in **hardware**.

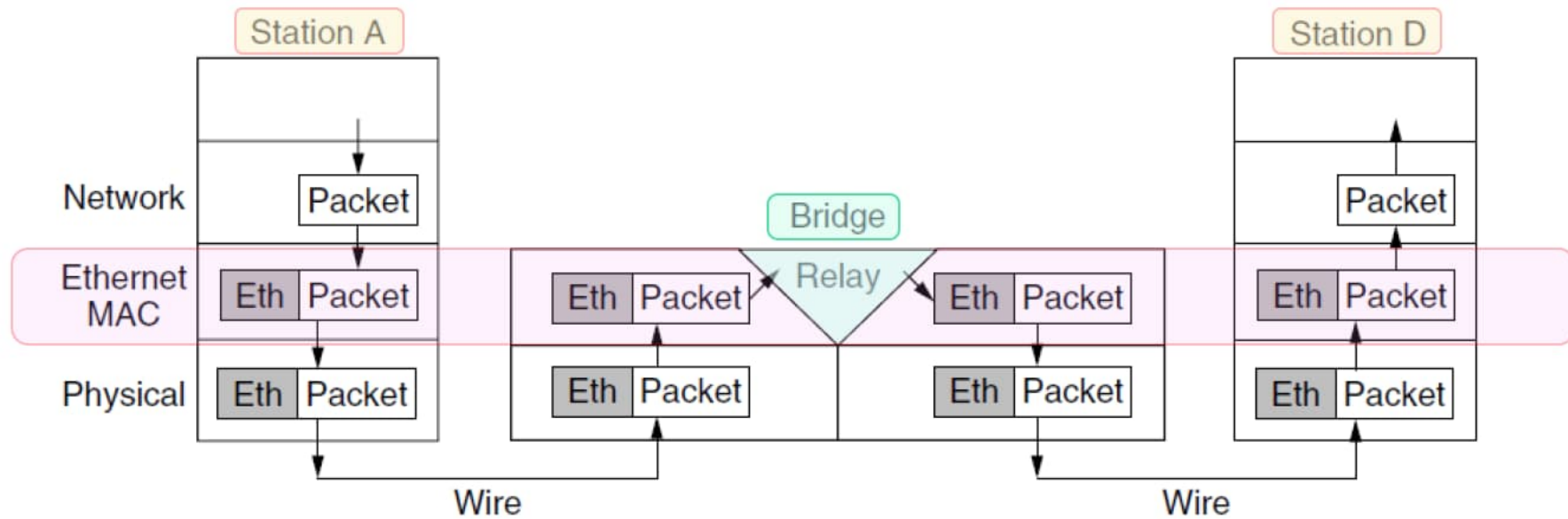


Figure 4-42. Protocol processing at a bridge.

4.8.3 Spanning Tree Bridges

➤ Spanning Tree Bridges 生成树网桥

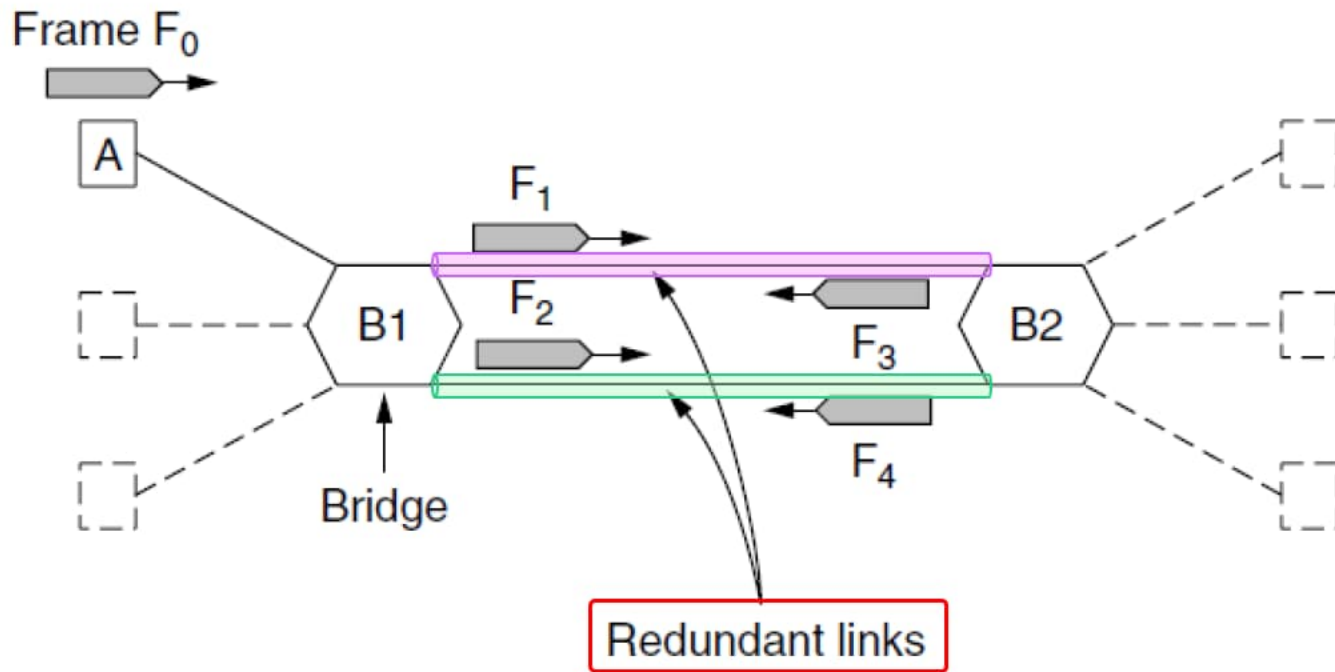


Figure 4-43. Bridges with two parallel links.

spanning tree

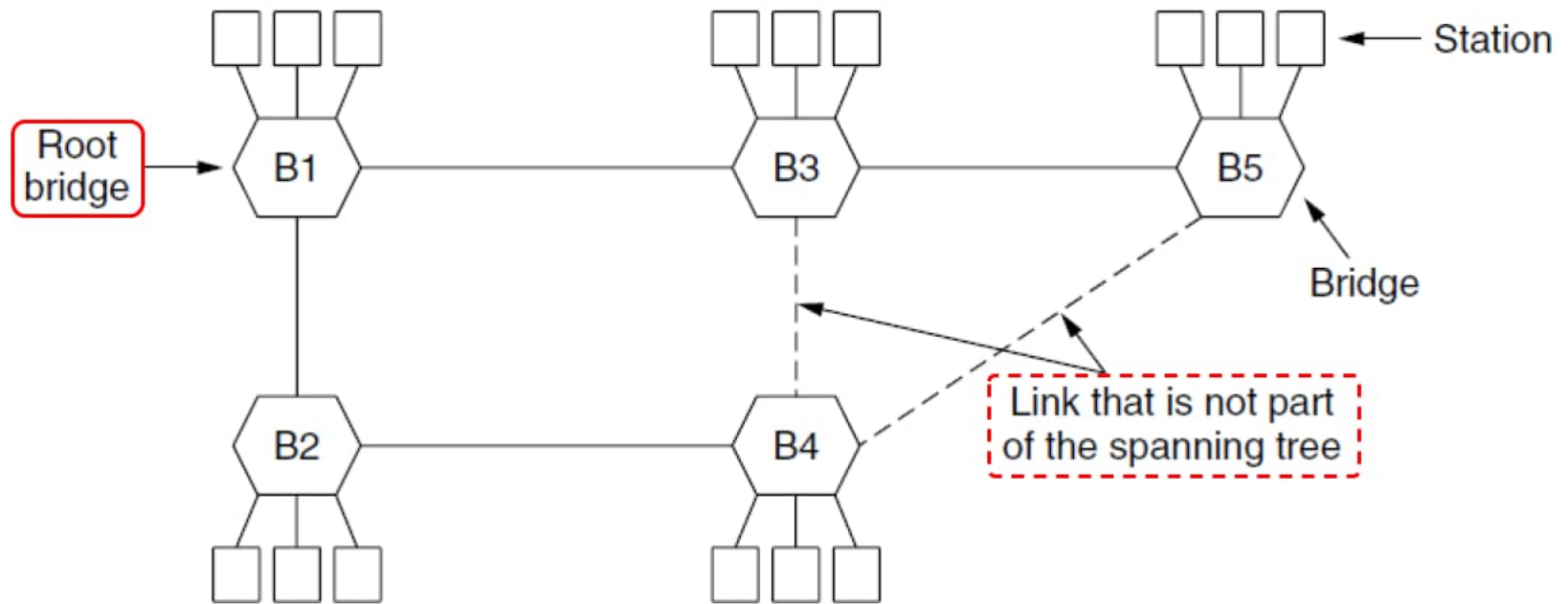


Figure 4-44. A spanning tree connecting five bridges.
The dashed lines are links that are not part of the spanning tree.

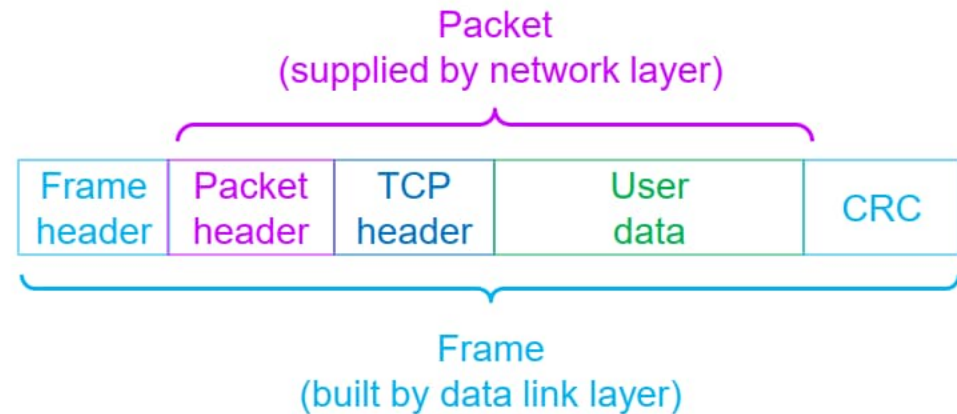
Radia Perlman's algorithm, 1985

- ✦ *I think that I shall never see*
- ✦ *A graph more lovely than a **tree**.*
- ✦ *A tree whose crucial property*
- ✦ *Is **loop-free connectivity**.*
- ✦ *A tree which must be sure to **span**.*
- ✦ *So packets can reach every LAN.*
- ✦ *First the **Root** must be selected*
- ✦ *By **ID** it is elected.*
- ✦ ***Least cost paths** from Root are traced*
- ✦ *In the tree these paths are placed.*
- ✦ *A mesh is made by folks like me*
- ✦ *Then bridges find a **spanning tree**.*

4.8.4 Repeaters, Hubs, Bridges, Switches, Routers, and Gateways

Application layer	Application gateway
Transport layer	Transport gateway
Network layer	Router
Data link layer	Bridge, switch
Physical layer	Repeater, hub

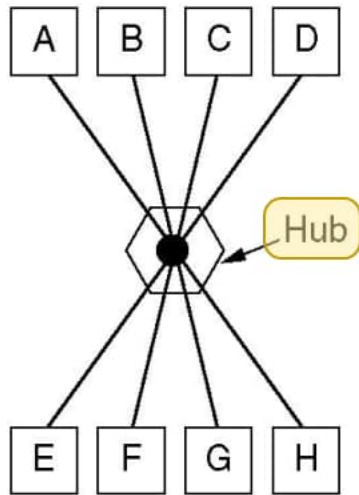
(a) Which device is in which layer.



(b) Frames, packets, and headers.

Figure 4-45.

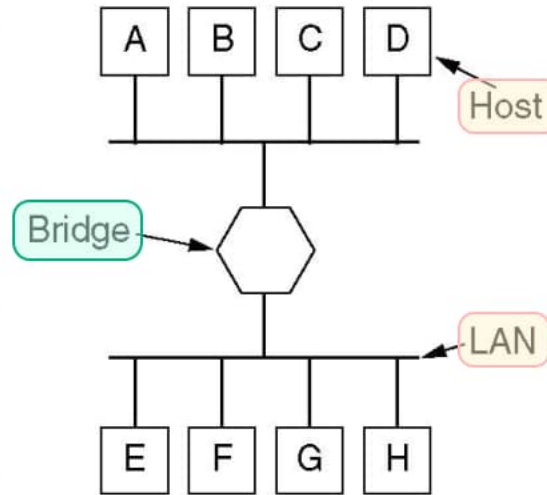
hub, bridge and switch



(a)

(a) A hub.

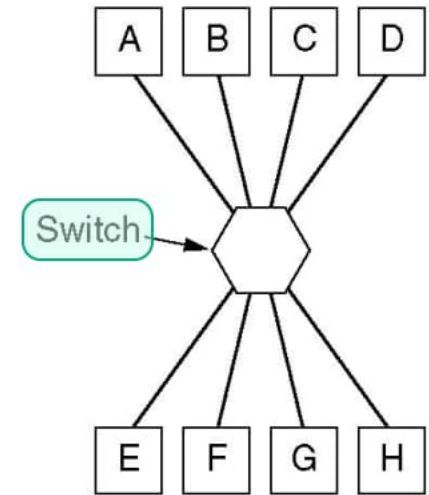
集线器



(b)

(b) A bridge.

网桥



(c)

(c) A switch.

交换机

Summary

- The key design issue is
 - ✦ the **allocation** of this channel **among the competing stations** wishing to use it.
- Dynamic channel allocation algorithms
 - ✦ ALOHA protocol
 - ✦ CSMA
 - 1-persistent
 - non-persistent
 - p-persistent
 - ✦ Collision-Free Protocols
 - ✦ Ethernet, CSMA/CD
 - ✦ IEEE 802.11, CSMA/CA

DLL: Data Link Layer

- LLC: Logical Link Control
- MAC: Medium Access Control

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Data Link

Chapter 2 Physical



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2. The Physical Layer
3. The Data Link Layer
4. The Medium Access Control Sub-Layer
5. The Network Layer
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8. Network Security

DLL: Data Link Layer

- LLC: Logical Link Control
- MAC: Medium Access Control

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Chapter 2 Physical

DLL

Summary



- The key design issue is
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